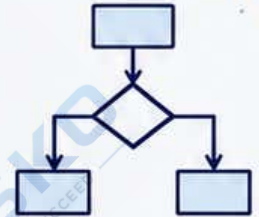


1010101010
0101010101
1010101010
0101010101



10110
01001
10101



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!

Most Important Questions

DATA STRUCTURES

ALGORITHMS

COMPUTER NETWORKS

DATABASE SYSTEMS

Same Goal
Brighter Future
Keep Going
You Can Do It!



OOPs with C ++ — Most Important Questions

Unit I - OOP Paradigm & Concepts

- **Answers in one or two lines (1 mark questions)**

1. What is OOP?
2. What is an OOP paradigm?
3. State any two features of OOP.
4. What is data abstraction?
5. What is encapsulation?
6. What is a class?
7. What is an object?
8. What is structured programming?
9. Name any two OOP languages.

- **Short Answer Questions (3 marks questions)**

1. Explain the basic concepts of OOP.
2. Explain the features of OOP.
3. Explain advantages and applications of OOP.
4. Differentiate between Structured Programming and OOP.
5. Explain data abstraction and encapsulation.
6. Explain classes and objects.

- **Long Answer Questions (6 marks questions)**

1. Explain OOP paradigm, its features, advantages and applications.
2. Differentiate between Structured Programming and Object-Oriented Programming.
3. Explain data abstraction, encapsulation, classes and objects with suitable examples.
4. Explain trending OOP languages and their applications.

Unit II - Functions & Constructors

- **Answers in one or two lines (1 mark questions)**

1. What is a function prototype?
2. What is an inline function?
3. What is a default argument?
4. What is function overloading?
5. What is a constructor?
6. What is a default constructor?
7. What is a parameterized constructor?
8. What is a copy constructor?

- **Short Answer Questions (3 marks questions)**

1. What is a Function prototype ? Explain function prototype with example.
2. What is an inline function ? Explain inline functions with syntax and examples.
3. What is the default argument ? Explain default arguments with syntax and examples.
4. What is Function overloading ? Explain function overloading with examples.
5. What is a Constructor ? Explain types of constructors.
6. Differentiate between default and parameterized constructors.
7. Explain copy constructor with example.

- **Long Answer Questions (6 marks questions)**

1. What is Function Overloading ? Explain function overloading with suitable examples.
2. Explain constructors and their types — Default, Parameterized and Copy Constructor.
3. Explain default, parameterized and copy constructors with suitable C++ programs.
4. Explain function prototype, inline function and default arguments.

Unit III - Functions, Objects & Inheritance

- **Answers in one or two lines (1 mark questions)**

1. What is an access specifier?
2. Name the access specifiers in C++.
3. What is memory allocation for objects?
4. What are objects as function arguments?
5. What is returning an object from a function?
6. What is inheritance?
7. Name the types of inheritance.
8. What is single inheritance?
9. What is multiple inheritance?
10. What is multilevel inheritance?

- **Short Answer Questions (3 marks questions)**

1. What are access specifiers ? Explain access specifiers in C++.
2. Explain memory allocation for objects.
3. Explain objects as function arguments.
4. Explain returning objects from functions.
5. What is Inheritance ? Explain inheritance and its types.
6. Explain single, multiple and multilevel inheritance.
7. Explain hierarchical and hybrid inheritance.
8. Explain visibility modes in inheritance.

- **Long Answer Questions (6 marks questions)**

1. Explain inheritance and all its types with suitable diagrams/examples.
2. Explain single, multiple, multilevel, hierarchical and hybrid inheritance.
3. Explain access specifiers and visibility modes in inheritance.
4. Explain objects as function arguments and returning objects from functions.

Unit IV - Polymorphism & Operator Overloading

- **Answers in one or two lines (1 mark questions)**

1. What is polymorphism?
2. What is compile-time polymorphism?
3. What is run-time polymorphism?
4. What is a virtual function?
5. What is a pure virtual function?
6. What is a virtual base class?
7. What is early binding?
8. What is late binding?
9. What is function overriding?
10. What is operator overloading?
11. What is unary operator overloading?
12. What is binary operator overloading?

- **Short Answer Questions (3 marks questions)**

1. What is Polymorphism ? Explain compile-time and run-time polymorphism.
2. Explain virtual functions.
3. Explain pure virtual functions.
4. Explain virtual base classes.
5. Differentiate between early binding and late binding.
6. Explain function overriding.
7. Explain operator overloading.
8. Explain unary and binary operator overloading.
9. State the rules for operator overloading.

- **Long Answer Questions (6 marks questions)**

1. Explain polymorphism and its types: Compile-time and Run-time.
2. Explain virtual functions and pure virtual functions with examples.
3. Explain early binding and late binding.
4. Explain operator overloading with unary and binary operators.
5. Explain function overriding and virtual functions with suitable examples.
6. Explain the rules for operator overloading.